

The Impacts of London's Ultra Low Emission Zone on Air Quality Using Machine Learning and Partially Pooled Synthetic Control Method

Ya-Chih Tsai (2852036)

supervisor: Prof. Dr. Julia Schaumburg

co-reader: Prof. Dr. Jan Magus



Abstract

Abstract

London’s Ultra Low Emission Zone (ULEZ) is considered the largest clean air intervention in the world. We apply a two-stage approach—machine learning-based meteorological normalization followed by a partially pooled synthetic control method (PPSCM)—to assess the impact of ULEZ (including major boundary expansions) on nitrogen dioxide (NO_2) and nitrogen oxide (NO_x) concentrations, using hourly monitoring data from 2018 to 2024. Average post-policy reductions of -5.4% for NO_2 and -6.4% for NO_x are found over the first 52 weeks following each cohort’s respective treatment date, although aggregate confidence intervals marginally include zero. Sub-period estimates are consistently negative across all time windows, and both spatial placebo tests and leave-one-out cross-validation confirm that these reductions are unlikely to arise from random spatial variation or the influence of any single monitoring site. This study advances the literature on air quality policy evaluation by adopting advanced meteorological normalization and using state-of-the-art causal inference with staggered adoption together with extensive falsification checks. The results emphasize not only the empirical complexity of evaluating large-scale urban interventions but also the promise that sustained emissions restrictions hold to improve urban air quality. The methodological framework developed here provides a reproducible blueprint for future multi-intervention environmental policy analysis.

1 Introduction

As cities around the world have industrialized, the public health risks associated with urban air pollution have become increasingly severe. Traffic emissions are among the leading contributors to dangerous levels of nitrogen dioxide (NO_2), particulate matter (PM_{10} and $\text{PM}_{2.5}$), ozone (O_3) and other pollutants in city centers, which are positively associated with cardiovascular, respiratory, and cerebrovascular mortality (Orellano et al. 2020), making regulation of vehicular access an essential policy tool. One widely adopted approach is the establishment of Low Emission Zones (LEZ), where vehicles are charged or restricted according to their age, fuel type, or EU emissions standard.

On 8 April 2019, London introduced one of the strictest Ultra Low Emission Zones (ULEZ) in the world within its existing Congestion Charge area (Mayor of London 2019). Originally limited to central London, the ULEZ applies to all passenger cars, motorcycles, light commercial vehicles, specialist vehicles, and minibusses that drive within its boundaries. Vehicles must meet Euro 4 emission standards for petrol cars, Euro 6 for diesel cars and Euro 3 for motorcycles, or pay a daily charge of £12.50, while busses and heavy-goods vehicles incur a £100 daily fee (Transport of London 2025). On 25 October 2021, the zone was extended north and south to the North and South Circular roads (Transport of London 2021). Finally, on 29 August 2023 it was further expanded to encompass the entire Greater London region (Greater London Authority 2023).

According to the Organization for Economic Cooperation and Development report (OECD, 2022), on an average day almost 94% of vehicles entering the ULEZ comply with emission standards, while compliance on the boundary roads and in the south west of London is 90% and 85%, respectively. Similarly, tightened LEZ standards for large heavy vehicles increased their compliance rate from 48% in 2017 to 96% in 2022 (OECD, 2022). In particular, since the implementation of ULEZ, there have been on average 44,000 fewer diesel vehicles on London’s roads each day—a reduction of approximately 20%—which not only slightly decreased overall vehicle numbers and traffic flows (by

approximately 2%), but also delivered significant improvements in air quality and public health (OECD, 2022). More broadly, since the introduction of the congestion charge, private car traffic in inner London initially decreased dramatically, but the freed-up road space was quickly filled by exempt taxis and other service vehicles. Although cars and taxis have driven approximately one billion fewer miles (about a 6% decrease) since 2000, light commercial vehicles have increased their mileage by a similar amount, leaving London’s total volume of road transport relatively unchanged at around 20 billion vehicle miles per year in the past two decades (OECD, 2022).

However, when researchers use different methods to evaluate the impact of London’s Ultra Low Emission Zone (ULEZ), the results are mixed (Hajmohammadi et al. 2022, Ma et al. 2021, Prieto-Rodriguez et al. 2022). This seems to stem from differences in data coverage and model assumptions, as well as the difficulty of separating the effects of ULEZ from other concurrent shocks, such as changes in weather or other traffic policies. Furthermore, most existing studies focus only on the first implementation of ULEZ within the congestion charging zone and ignore the second and third expansions—therefore, they cannot track the impact of the policy towards the whole of Greater London.

The purpose of this paper is to employ data-driven techniques—specifically machine learning and synthetic control methods—to estimate the impact of the ultra-low emission zone (ULEZ) on London’s air quality, as recorded by fixed monitoring stations between 2018 and 2024. We aim to measure how the introduction and subsequent expansions of ULEZ have influenced the concentrations of key pollutants such as NO_2 and NO_x .

To achieve this, we combine a machine learning–based meteorological normalization that filters out weather and seasonal influences, and then apply the synthetic control method as our main analysis method to estimate the impact of the ULEZ’s policy. Drawing on seven years of hourly pollution data from London’s 13 air quality stations along with meteorological records from the European Center for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis dataset with surface observations sup-

plemented by the National Oceanic and Atmospheric Administration (NOAA) , we estimate the independent staggered effects of the three boundary extensions. Our analysis delivers a robust assessment of how each ULEZ expansion altered traffic-related pollutant levels both in central London and in the Greater London area, offering insights that can guide the design and evaluation of clean-air policies worldwide.

Evaluating large-scale policies such as the ULEZ is highly challenging, because air pollution results from a complicated interaction of several factors: meteorological conditions (wind direction, wind speed, air pressure, precipitation, temperature, humidity, etc.), traffic volume, economic policies, social trends, and COVID-19 all have influences on the levels of pollution. The methodology of our study uses data-driven algorithms to overcome these challenges and therefore provides a reproducible framework for policy evaluation.

This study has the following contributions: it is the first to apply the Synthetic Control Method to evaluate the London ULEZ policy, offering an alternative analytical perspective to existing research. Second, it demonstrates how a two-step framework—meteorological normalization followed by a synthetic control method with staggered adoptions - offers insight into analyzing policies implemented at different points in time. Together, it gives a well-checked, easy-to-copy plan for evaluating environmental policies in cities around the globe like ULEZ.

This thesis is divided into five chapters. Chapter 1 provides the introduction, outlining the research background and objectives. Chapter 2 reviews the existing literature, summarizing the methods previously employed and the conclusions they reached. Chapter 3 introduces the data used in this study. Chapter 4 describes models, estimation techniques, and testing procedures. Chapter 5 presents the empirical analysis, including the main results and a series of robustness checks, while Chapter 6 provides conclusions and discussions.

2 Literature Review

Existing studies have used data sets with varying temporal coverage and applied different analytical methods to evaluate the effects of the Ultra Low Emission Zone (ULEZ). This chapter reviews the most relevant work, briefly describing the data they used, the methods they applied, and what they found. Finally, by comparing those studies to the approach used here, we show what our research adds.

Bornioli et al. (2022) analyzed 16 air quality monitoring stations within the ULEZ, comparing spring 2018 with spring 2019 (8 April–7 July). They first examined trends from 2014 to 2018 and found that NO_2 was already falling. Then they used paired t tests and Wilcoxon signed rank tests to show that in eight of the nine inner-zone sites NO_2 decreased significantly after ULEZ started, while PM_{10} only fell at three sites and by a small amount. By matching the same season each year and including four outer-zone sites as controls, they tried to rule out weather and seasonal effects. However, they did not include actual meteorological data in their models, only seasonal pairing, so their results still carry some uncertainty. They also only looked at the first 90 days after ULEZ began, missing longer-term effects, and mixed roadside and background stations in ways that could overstate or understate real changes.

Liang Ma et al. (2021) used hourly data from 79 roadside and background stations throughout Greater London, covering January 1, 2016 to January 31, 2020. They trained a machine-learning model using temperature, wind speed and direction, humidity, rainfall, time of day, and seasonal factors to predict pollutant levels in the absence of ULEZ. This “weather-normalized” series removed meteorological and seasonal patterns. They then applied change-point detection around 8 April 2019 and implemented a donut Regression Discontinuity Design (RDD) to compare pollutant levels before and after the introduction of ULEZ. Within five to eight weeks, approximately 70% of the stations recorded significant drops in NO_2 and 50% in NO_x , while only 24% showed changes in O_3 . On average, NO_2 and NO_x decreased by roughly 3%, while O_3 and $\text{PM}_{2.5}$ remained largely unchanged. This indicates that ULEZ induced rapid

local reductions but was insufficient for sustained, citywide improvement—highlighting the need for broader multi-zone policies. However, since sharp RDD estimates effects only at the cut-off point, it cannot capture longer-term dynamics and is susceptible to coincident shocks.

Hajmohammadi et al. (2022) were the first to apply a State-Space model to assess the impacts of ULEZ’s on NO, NO₂, and NO_x. Using hourly data from 28 stations that span 12 months before and 11 months after ULEZ began, they constructed a log-linear model incorporating temperature, humidity, wind speed, and temporal cycles (monthly and weekly). Their results indicated that within the ULEZ, NO and NO_x decreased by approximately 19–20% and NO₂ decreased by roughly 11.6%. In the Greater London Area, NO_x decreased by 15–18%. By 2020, the annual average NO₂ (38.9 µg/m³) fell below the EU limit. Seasonal patterns were evident, with lower pollutants from June to October and on weekends. However, their model assumes linear, time-invariant relationships among stations and treats ULEZ as an abrupt, constant shock, overlooking gradual behavioral change or episodic events such as roadworks. It also presumes white noise errors, whereas systematic measurement bias and cross-site dependence likely persist.

Prieto-Rodríguez et al. (2022) examined NO₂ levels at 13 treated sites within the ULEZ and 16 control sites more than 15 km away using daily averages from 1 January 2015 to 29 February 2020. Their DiD model controlling for meteorology and year, month and weekday fixed effects reported that after the introduction of ULEZ on 8 April 2019 was applied, roadside sites decreased in NO₂ by 14.9 µg/m³ (19%) and background sites by 7.0 µg/m³ (19.3%). NO levels 2–5 km outside the border decreased by almost 11 µg/m³, indicating that there were spillover effects. However, the assumption of parallel trends is not formally validated in the study—a necessary condition for identification in DiD—and the graphical evidence is based only on visual inspection of disaggregated trends without conducting statistical tests. In addition, the lack of robustness checks such as placebo tests, event study specifications, or alternative control group definitions undermines confidence in the estimated effects, especially since fixed effects can only

control for time-invariant attributes and do not allow for time-varying confounders.

To offer different angles of analysis and interpretation, we adopt the Synthetic Control Method (SCM) introduced by Abadie et al. (2010), which has been described as “the most important innovation” in the literature on policy evaluation over the past 15 years (Athey and Imbens. 2017, p. 12). SCM is particularly appropriate for evaluating interventions or events that occur at a fairly aggregate level, such as interventions that involve cities, regions, and countries in their economic and social outcomes. In addition, SCM can capture the medium- and long-term effects of a policy more effectively (Abadie. 2021). However, given that traditional SCM requires a uniform treatment timing in all units, this study adopts Partially Pooled SCM (Ben-Michael et al.) to accommodate settings with a staggered treatment adoption (Ben-Michael et al 2022).

Therefore, in this study, we first apply a weather normalization model to pollutant concentration data before and after London’s ULEZ. and then adopt a synthetic control method to de-weathered concentrations to estimate the causal impact of ULEZ.

3 Data

We used hourly concentrations of NO_2 and NO_x obtained from 56 air quality monitoring stations in the United Kingdom. Data were collected through the open-source R package `openair` (Carslaw and Ropkins, 2012), which retrieves air quality data from the UK Department of Environment, Food & Rural Affairs (DEFRA) monitoring network. Specifically, the treatment group comprises 13 stations in Greater London that are part of the King’s College London (KCL) monitoring network, whose station types include Kerbside, Roadside, Urban Traffic, Urban Background and Suburban sites. The remaining 38 stations, which serve as the control group, are from DEFRA’s UK Automatic Urban and Rural Network (AURN), located in other UK cities to construct counterfactual levels of air pollution for Greater London.

The air quality data from DEFRA AURN were accessed through the DEFRA Air Quality Archive (<https://uk-air.defra.gov.uk/data>) and downloaded on 15 June 2025,

using the `openair` function `importAURN()`. The KCL network data were obtained from the London Air Quality Network managed by King’s College London Environmental Research Group (<https://www.londonair.org.uk/>) through the `openair` function `importKCL()` on the same date.

The London Ultra Low Emission Zone (ULEZ) policy was imposed in three stages: initially on 8 April 2019, within the Central London congestion charge zone; expanded on 25 October 2021, covering all areas within the North and South Circular roads; and finally extended on 29 August 2023, to cover the entire Greater London administrative area. Our evaluation period spans January 1, 2018 to December 31, 2024, covering before and after policy implementation.

Meteorological data corresponding to the study period were collected hourly. The primary source was the European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis dataset (<https://doi.org/10.24381/cds.adbb2d47>) supplemented by surface observations from National Oceanic and Atmospheric Administration (NOAA) airport weather stations, accessed via the `importNOAA()` function in the R package `worldmet` (Carslaw and Davison, 2025). Meteorological variables included wind speed and direction, air temperature, relative humidity, surface pressure, boundary layer height, surface net solar radiation, total cloud cover, and total precipitation. Temporal variables derived from timestamps—such as Unix date, week number, day of week, hour of day, month, and Julian day—were also used as predictors in further analyses. The meteorological data were downloaded on July 12, 2025.

A two-stage approach was used to estimate the causal effect of ULEZ on air quality in Greater London. The first stage uses machine learning-based meteorological normalization techniques (Grange et al. 2018; Grange et al. 2019) to decompose weather influences from observed pollutant concentrations, producing normalized concentrations indicative of actual emission changes. The second stage applies the Partially Pooled Synthetic Control Method (Ben-Michael et al. 2022) using meteorologically normalized concentration data to create synthetic control units and estimate the causal effect of ULEZ.

Table 1: Descriptive Statistics for Treated Group and Control Group (2018–2024)

Variable	Treated Group (London)			Control Group (UK Cities)		
	Mean (SD)	Median	Range	Mean (SD)	Median	Range
<i>Primary Pollutants</i>						
NO ₂ (μg/m ³)	27.65 (18.66)	24.00	[0.00, 319.50]	21.88 (17.86)	16.69	[−0.38, 409.99]
Missing (%)		6.40			–	
NO _x (μg/m ³)	52.61 (61.37)	35.30	[0.00, 1689.70]	42.06 (54.43)	23.91	[0.00, 1507.81]
Missing (%)		6.59			–	
<i>Meteorological Variables</i>						
Wind Speed (m/s)	3.96 (2.13)	3.60	[0.00, 18.20]	4.64 (2.40)	4.29	[0.00, 20.61]
Wind Direction (°)	192.53 (94.02)	210.00	[0.00, 360.00]	199.03 (92.60)	218.89	[0.00, 360.00]
Air Temperature (°C)	12.06 (6.13)	11.60	[−6.10, 39.10]	10.87 (5.51)	10.66	[−8.21, 39.15]
Relative Humidity (%)	74.96 (16.27)	78.00	[13.00, 100.00]	81.04 (12.80)	84.07	[14.52, 100.01]
Atmospheric Pressure (hPa)	1016.32 (11.08)	1017.20	[958.40, 1049.50]	1003.59 (12.64)	1004.59	[945.26, 1045.08]
Solar Radiation (W/m ²)	106.19 (165.64)	4.00	[0.00, 896.00]	102.65 (160.96)	4.28	[0.00, 891.00]
Boundary Layer Height (m)	645.72 (449.61)	573.00	[10.00, 2901.00]	639.02 (410.95)	590.29	[9.76, 4536.97]
Cloud Cover (%)	68.61 (35.79)	88.00	[0.00, 100.00]	69.71 (34.19)	86.26	[0.00, 100.00]
Precipitation (mm)	0.08 (0.32)	0.00	[0.00, 14.30]	0.11 (0.33)	0.00	[0.00, 10.61]

Note: The treated group consists of 18 monitoring sites in London (N = 1,104,624 hourly observations). The control group consists of 38 monitoring sites across 19 UK cities. The study period covers 2018-01-01 to 2024-12-31. Meteorological data are sourced from city-specific weather datasets.

Table 2: Treatment Group Stations in Greater London

Code	Borough	Location	Site Type	Zone
MY1	Westminster	Marylebone Road	Kerbside	Central
WM6	Westminster	Oxford Street	Urban Traffic	Central
SK6	Southwark	Elephant & Castle	Roadside	Central
WM5	Westminster	Covent Garden	Urban Background	Central
IS2	Islington	Holloway Road	Roadside	Inner
RI1	Richmond	Castelnau	Urban Traffic	Inner
TH2	Tower Hamlets	Mile End Road	Roadside	Inner
KC1	Kensington	North Ken	Urban Background	Inner
BT8	Brent	ARK Franklin	Roadside	Inner
LW4	Lewisham	Loampit Vale	Roadside	Inner
TH4	Tower Hamlets	Blackwall	Roadside	Inner
EN1	Enfield	Bush Hill Park	Suburban	Outer
CR5	Croydon	Norbury	Kerbside	Outer
HV3	Havering	Romford	Roadside	Outer
GN3	Greenwich	Plumstead High St.	Roadside	Outer
EA6	Ealing	Hanger Lane	Roadside	Outer
EI1	Ealing	Western Avenue	Roadside	Outer
EA8	Ealing	Horn Lane	Roadside	Outer
<i>Total: 18 monitoring stations (4 Central, 7 Inner, 7 Outer)</i>				

4 Methodology

4.1 Random Forest Model

Grange et al. (2018, 2019) demonstrate that the random forest model effectively addresses the issues of multicollinearity and interaction effects. As a data-driven, non-parametric machine learning approach, it imposes no assumptions regarding the underlying data structure, rendering it highly applicable to diverse air quality intervention studies. We developed individual models for each pollutant and monitoring site, allowing a more precise and adaptable adjustment to meteorological influences.

The fundamental principle of this methodology involves employing the random forest model to isolate and remove the effects of meteorological variability on pollutant concentrations, thus uncovering authentic trends attributable to emissions or chemical transformations. Specifically, when the random forest model accounts for a substantial portion of the variance in the dependent variable, repeated resampling and prediction procedures can effectively eliminate the influence of explanatory meteorological variables on the outcome. This study seeks to evaluate the effects on emissions after the implementation of London’s Ultra Low Emission Zone (ULEZ) policy, drawing methodological inspiration from the causal inference framework utilized by Liu et al. (2023) in their analysis of Birmingham’s Clean Air Zone (CAZ) policy.

For each monitoring site s and pollutant p , the random forest regression model is formally specified as:

$$C_{s,p,t} = f(X_{met,t}, X_{temp,t}) + \epsilon_t \quad (1)$$

where:

Dependent variable: $C_{s,p,t}$ represents the concentration of pollutant ($p \in \{\text{NO}_2, \text{NO}_x\}$) at the site s and time t (hourly observations).

Meteorological features ($X_{met,t}$): Wind speed, wind direction, air temperature, relative humidity, atmospheric pressure, solar radiation, boundary layer height, cloud cover

and precipitation.

Temporal features ($X_{temp,t}$): Date_unix (seconds since 1970-01-01), day_julian (1-365), week (1-52), weekday (Monday-Sunday), hour (0-23) and month (1-12).

The model function: $f(\cdot)$ represents the random forest ensemble consisting of B decision trees, with predictions obtained as:

$$\hat{C}_{s,p,t} = \frac{1}{B} \sum_{b=1}^B T_b(X_{met,t}, X_{temp,t}) \quad (2)$$

where T_b denotes the b -th decision tree trained on a bootstrap sample

Following existing studies by Grange et al (2019) and Liu et al. (2023), the ensemble consists of decision trees $B = 500$, and at each node split during tree construction, the algorithm randomly samples $m_{try} = 5$ variables as split candidates. And a minimum node size of 5 observations.

Initially, a predictive random forest model was constructed that incorporated meteorological and temporal predictors as outlined previously. Data spanning January 1, 2018 to December 31, 2024, were used to forecast hourly concentrations of nitrogen dioxide (NO_2) and nitrogen oxides (NO_x) at 50 air quality monitoring stations throughout the United Kingdom.

Model performance was evaluated using out-of-bag (OOB) estimates. The coefficient of determination is computed as follows:

$$R_{OOB}^2 = 1 - \frac{\sum_{i \in OOB} (C_i - \hat{C}_i)^2}{\sum_{i \in OOB} (C_i - \bar{C})^2} \quad (3)$$

where C_i are observed concentrations, $\hat{C}_i$ are OOB predictions and $\bar{C}$ is the mean observed concentration.

In the context of meteorological normalization, concerns about overfitting are different from those of conventional predictive modeling. Our primary objective is not to forecast future pollutant concentrations, but rather to characterize the functional relationship between meteorological conditions and observed pollution levels. To address potential overfitting issues, we adopted the bootstrap aggregating (bagging) framework

by Grange and Carslaw (2019). During model training, each decision tree is grown using a bootstrap sample drawn with replacement from the original data set, automatically generating out-of-bag (OOB) observations—approximately one-third of the data excluded from each tree’s training. These OOB samples effectively serve as an independent validation set, providing unbiased estimates of model performance.

4.2 Weather Normalization Procedure

Meteorological normalization was conducted independently for each pollutant at each monitoring location after model training and validation. For example, at the Westminster–Marylebone Road site, a specific time point (6:00 a.m. on April 14, 2019). The pre-trained and validated random forest model was then used to generate 500 predictions of NO_2 concentration at this time and site. In each iteration, the meteorological variables were randomly resampled without replacement throughout the study period (1 January 2018 to 31 December 2024) and entered into the model. The arithmetic mean of these 500 predictions was calculated to represent the meteorologically normalized (de-weathered) NO_2 concentration for the specified time and location. An identical procedure was applied to NO_x using the same computational framework, resulting in hourly meteorologically normalized concentrations for all sites throughout the duration of the study. Subsequently, both observed and normalized hourly concentrations were aggregated into weekly averages to facilitate the second-stage analysis.

4.3 Partially Pooled Synthetic Control Method

Meteorological normalization removes weather-driven variation from pollutant concentrations, but simply comparing pre- and post-ULEZ normalized series is insufficient (Shi et al. 2021). We therefore adopt the synthetic control method. In brief, SCM constructs a weighted average of control units whose pre-treatment trajectory closely matches that of the treated unit and uses this synthetic to approximate the absence of counterfactual policy (Abadie et al. 2010, 2015, 2021).

However, traditional SCM assumes that all units receive treatment simultaneously, which is not consistent with our case. Our setting features staggered adoption: Central London (April 2019), Inner London (October 2021), and Outer London (August 2023).

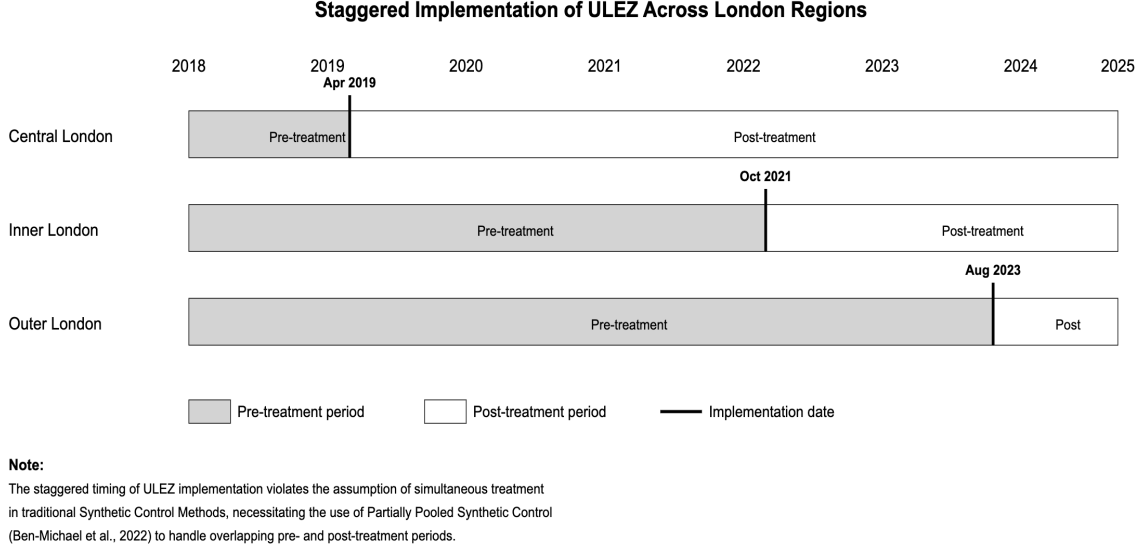


Figure 1: Staggered Implementation of ULEZ Across London Regions. The figure illustrates the timing of Ultra Low Emission Zone implementation across three London regions.

To handle staggered timing, we use the Partially Pooled Synthetic Control by Ben-Michael et al. (2022), which jointly targets two pre-treatment imbalance measures in a single optimization problem. The unit-specific fit $q_{\text{sep}}(\Gamma)$ measures the root mean squared error (RMSE) between each treated unit's pre-treatment outcomes and its synthetic control, while the pooled fit $q_{\text{pool}}(\Gamma)$ measures the RMSE for the average across all treated units:

$$\min_{\Gamma \in \Delta^{\text{scm}}} \nu \left(\tilde{q}_{\text{pool}}(\Gamma) \right)^2 + (1 - \nu) \left(\tilde{q}_{\text{sep}}(\Gamma) \right)^2 + \lambda \|\Gamma\|_F^2, \quad (4)$$

where $\Gamma \in \mathbb{R}^{N \times J}$ is the weight matrix with element γ_{ij} denoting the weight on donor unit i for treated unit j . The constraint set Δ^{scm} requires: (i) non-negativity $\gamma_{ij} \geq 0$, (ii) summation to unity $\sum_i \gamma_{ij} = 1$, and (iii) $\gamma_{ij} = 0$ for invalid donors $i \notin D_{jK}$, where $D_{jK} = \{i : T_i > T_j + K\}$ includes only units not yet treated by event time K relative

to unit j 's treatment date.

The hyperparameter $\nu \in [0, 1]$ balances the two objectives: $\nu = 0$ produces a separate SCM (minimizing only the unit-specific fit), $\nu = 1$ produces a pooled SCM (minimizing only the average fit), and $0 < \nu < 1$ produces a partial pooling solution. To achieve the optimal pre-treatment fit, we followed the recommendations of Ben-Michael et al. (2022) and used a balance frontier sensitivity analysis (q_{pool} versus q_{sep}) to determine the appropriate partial pooling parameter ν . Following Ben-Michael et al. (2022), ν is selected by inspecting the balance possibility frontier, which traces the Pareto-optimal trade-off between the pooled imbalance (q_{pool}) and the unit-specific imbalance (q_{sep}) across values of $\nu \in [0, 1]$. The recommended value is the point at which marginal improvements in q_{pool} become small relative to the additional unit-specific imbalance incurred, that is, the region of diminishing returns on the frontier. Based on this criterion, $\nu = 0.35$ is selected for NO_2 and $\nu = 0.25$ for NO_x . At these values, the pre-treatment fit metrics (RMSE and scaled imbalance L_2) remain well below the estimated magnitude of the treatment effect, satisfying the convex hull condition required for valid synthetic control inference (Abadie 2021). The penalty term $\lambda \|\Gamma\|_F^2$ serves two purposes: ensuring uniqueness when perfect balance is achievable, and improving numerical stability (Abadie et al. 2015). In our staggered adoption setting, Ben-Michael et al. (2022) note that a perfect balance between all treated units is typically infeasible, so uniqueness is guaranteed even without penalty. We set $\lambda = 10^{-6}$ primarily for numerical stability.

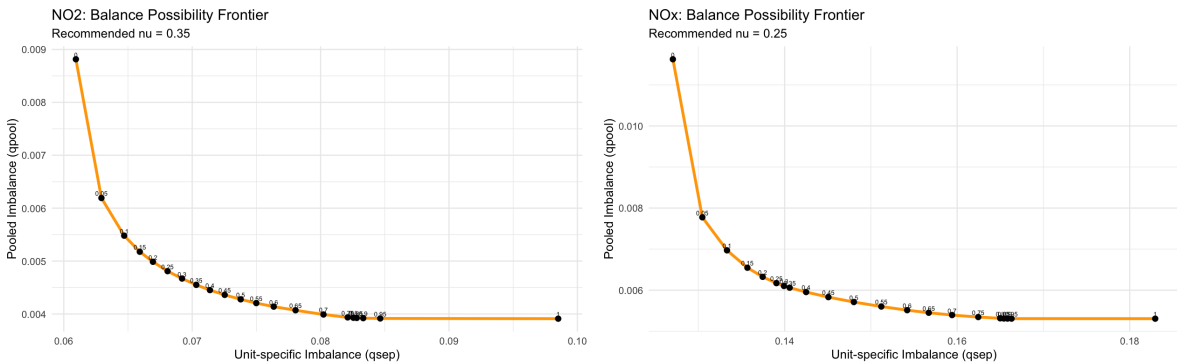


Figure 2: Balance frontier sensitivity analysis for NO_2 (left) and NO_x (right).

Given the estimated weights $\hat{\Gamma}$, the effect of treatment for unit j at the event time k is estimated as:

$$\hat{\tau}_{jk} = Y_{j,T_j+k} - \sum_{i=1}^N \hat{\gamma}_{ij} Y_{i,T_j+k}, \quad (5)$$

where $\sum_{i=1}^N \hat{\gamma}_{ij} Y_{i,T_j+k}$ is the synthetic control estimate of the counterfactual outcome $Y_{j,T_j+k}(\infty)$. The average effect of treatment on the treated at the event time k (periods relative to its own treatment date) is then:

$$\widehat{ATT}_k = \frac{1}{J} \sum_{j=1}^J \hat{\tau}_{jk}, \quad (6)$$

which is both the mean of unit-level effects and, equivalently, the effect for the average treated path.

In addition, site-level fixed differences are large in our data, we use the intercept-adjusted version of PPSCM, which is algebraically equivalent to a weighted difference-in-differences on de-meaned outcomes:

$$\hat{\tau}_{jk}^* = \frac{1}{L_j} \sum_{\ell=1}^{L_j} \left[(Y_{j,T_j+k} - Y_{j,T_j-\ell}) - \sum_{i=1}^N \hat{\gamma}_{ij}^* (Y_{i,T_j+k} - Y_{i,T_j-\ell}) \right], \quad \widehat{ATT}_k^* = \frac{1}{J} \sum_{j=1}^J \hat{\tau}_{jk}^*. \quad (7)$$

At each event time k , $\widehat{ATT}(k)$ represents the *average effect in the city (Greater London)*: the mean change for the treated sites k periods after their *own* adoption, relative to what those sites would have experienced in the absence of ULEZ (as approximated by their synthetic controls).

4.4 Assumptions and Validation Strategy

Abadie (2021) highlights that the validity of the synthetic control method is based on several assumptions, which have been addressed in this study through design choices. First, the no interference assumption (SUTVA) requires that the control units within the donor pool should not be influenced by the treatment policy, nor should there be any interference among the control units themselves. To meet this requirement, the

donor pool exclusively includes monitoring stations located outside Greater London that have never been subject to policy interventions and excludes all monitoring sites within Greater London. This ensures that units potentially impacted by the ULEZ policy, either simultaneously or prior to the study period, are omitted from the donor pool. Furthermore, all control stations remain completely unaffected by ULEZ or similar policies throughout the study period. To address possible spillover effects on surrounding areas caused by the ULEZ policy, monitoring sites near London that could be indirectly affected have also been excluded, further strengthening compliance with the no interference assumption. Secondly, the effectiveness of a synthetic control is based on the characteristics of the treated unit aligned with or close to the convex hull of the characteristics of the donor pool—a condition referred to as the convex hull condition. This study evaluates adherence to this condition by assessing whether the pre-treatment RMSE (root mean squared error) is significantly smaller than the estimated treatment effect. A considerably lower RMSE compared to the treatment effect confirms that the synthetic control can accurately mirror the pre-treatment trajectory of the treated unit using a weighted combination of units from the donor pool. Additionally, improvements in L2 imbalance after partial pooling optimization offer additional evidence to support conformity to the convex hull condition. Lastly, the synthetic control method presumes that the treated unit does not anticipate or adjust its behavior in response to the policy prior to its implementation date. This assumption, referred to as the no anticipation assumption, is tested in this study through multiple time placebo tests. These tests involve setting false policy dates 10, 15, 20, and 25 weeks before the actual implementation date to verify whether parallel trends are maintained between treated and synthetic control units prior to policy implementation. In addition, spatial placebo tests are performed using 200 random permutations to determine if the observed policy impacts could be merely attributed to random spatial variation. The detailed results of these robustness checks will be provided in Chapter 5.

5 Results

5.1 Weather Normalized Air Pollution Concentrations

Before presenting the main causal estimates, it is necessary to report the meteorological normalization results for the monitored and control sites during the period January 1, 2018 to December 31, 2024. Here, we focus primarily on the treated sites as examples.

Applying site-specific random-forest meteorological normalization to hourly NO_2 and NO_x data and aggregating to weekly resolution reveals a consistent seasonal cycle: concentrations are higher in winter and lower in summer.

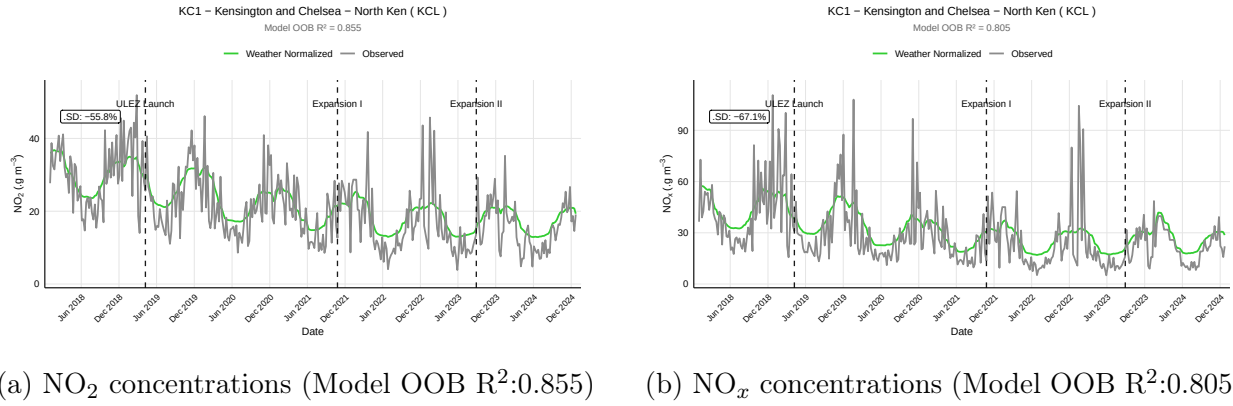


Figure 3: Weather-normalized and observed NO_2 and NO_x concentrations at Kensington and Chelsea monitoring - North Ken (KC1) site from 2018 to 2024. Green lines show weather-normalized values after removing meteorological influences. Black dash lines vertical lines mark ULEZ implementation phases.

This cycle persists after normalization, not because of residual meteorological noise but because temporal variables (such as Julian day, weekday, and hour of day) in the model act as proxies for emission activity and photochemical cycles; in contrast, short-term meteorological fluctuations have been effectively removed. As Grange et al. (2019) stated, the purpose of meteorological normalization is not to flatten the series, but to reduce meteorologically driven variability so that the underlying trends and intervention-related features become visible under “average” weather conditions. With only routine surface meteorology and time proxies, the method provides sufficient explanatory power, and the resulting normalized series are well-suited for subsequent

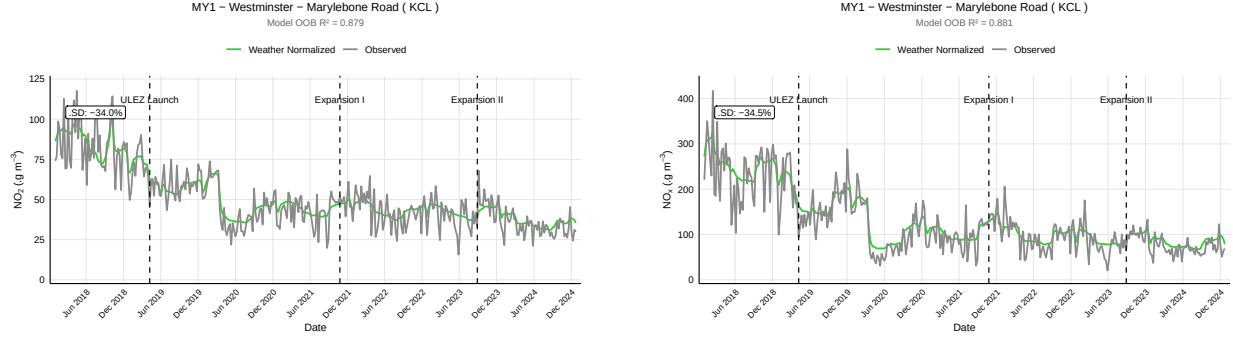
trend analysis or intervention detection, since policy signals often obscured by weather variation in raw data can become clearer once meteorological influences are removed.

In terms of model performance, based on the 18 treated sites, NO_2 models achieved out-of-bag (OOB) R^2 values between 0.79 and 0.879, while NO_x models ranged from 0.771 to 0.881. All remaining sites fell within these intervals. These levels of predictive capacity are consistent with those reported by Grange and Carslaw (2019), for example, the Marylebone Road site of Westminster in London (MY1; $\text{NO}_2 \approx 0.82$, $\text{NO}_x \approx 0.83$), providing external validation for the present normalization procedure.

To evaluate the degree of variance reduction, the standard deviation change rate (ΔSD) was calculated for each site. The results are predominantly large and negative, indicating that the normalized series are substantially less volatile than the observed ones, and that the short-term fluctuations were primarily meteorologically driven and successfully abstracted by the model. At the same time, because only meteorological variables were randomized during resampling—while time proxyes were preserved—the characteristic winter–summer cycle remains clearly visible. This is evidence of structural emission and chemical processes, rather than meteorological noise.

Marked heterogeneity between sites is also evident. KC1: NO_2 OOB $R^2 = 0.85$, $\text{NO}_x = 0.803$; MY1: $\text{NO}_2 = 0.882$, $\text{NO}_x = 0.886$), but their ΔSD values differ markedly: KC1 shows larger variance reductions ($\text{NO}_2 -55.8\%$, $\text{NO}_x -67.1\%$), while the reductions of MY1 are smaller ($\text{NO}_2 -34\%$, $\text{NO}_x -34.5\%$).

Other treated sites fall largely between these two: while they generally show both seasonality and long-term downward trends after normalization, visual inspection alone is insufficient to identify causal effects. This observation is consistent with the findings in Grange and Carslaw (2019), normalization improves visibility of potential intervention signals, but robust causal inference ultimately requires a formal framework.



(a) NO_2 concentrations (Model OOB R^2 :0.879) (b) NO_x concentrations (Model OOB R^2 :0.881)

Figure 4: Weather-normalized and observed NO_2 and NO_x concentrations at Westminster - Marylebone Road (MY1) site from 2018 to 2024. Green lines show weather-normalized values after removing meteorological influences. Black dash lines vertical lines mark ULEZ implementation phases.

5.2 Changes in Air Quality Relative to Partially Pooled SCM

The analysis reveals consistent patterns of pollutant reduction after ULEZ implementation, focusing on the policy-relevant window of 0 to 52 weeks post-treatment. Table 3 presents the summary results for both NO_2 and NO_x .

Table 3: ULEZ Treatment Effects and Model Diagnostics

	NO ₂			NO _x		
	ATT (%)	SE	95% CI	ATT (%)	SE	95% CI
Panel A: Overall Treatment Effects (0–52 weeks)						
Mean Effect	−5.4	0.029	[−10.5, +0.3]	−6.4	0.039	[−13.1, +1.2]
Panel B: Dynamic Effects by Time Window						
Short-term (0–15 weeks)	−3.9	0.005	[−4.8, −2.9]	−8.0	0.006	[−9.2, −6.9]
Mid-term (16–30 weeks)	−6.5	0.005	[−7.4, −5.6]	−7.9	0.004	[−8.7, −7.1]
Long-term (31–52 weeks)	−5.5	0.004	[−6.3, −4.7]	−3.5	0.009	[−5.1, −1.9]
Panel C: Spatial Heterogeneity by Cohort						
Central London (2019, <i>n</i> = 4)	−4.5	0.053	[−13.9, +5.9]	−12.2	0.035	[−17.9, −6.0]
Inner London (2021, <i>n</i> = 7)	−5.5	0.021	[−9.3, −1.4]	−4.8	0.038	[−11.6, +2.6]
Outer London (2023, <i>n</i> = 7)	−5.5	0.016	[−8.5, −2.4]	−4.1	0.022	[−8.1, +0.2]
Panel D: Model Diagnostics						
Scaled Global L ₂ Imbalance	0.020			0.022		
Scaled Individual L ₂ Imbalance	0.103			0.148		
Pre-treatment RMSE (log scale)	0.0146			0.0211		
Global Improvement (%)	98.0			97.8		
Individual Improvement (%)	89.7			85.2		
Average Effective Donors	3.83			4.17		
Optimal <i>ν</i> Parameter	0.35			0.25		
Panel E: Station-Level Fit Quality						
Excellent (<5% RMSE)	6 (33%)			2 (11%)		
Good (5–10% RMSE)	10 (56%)			8 (44%)		
Acceptable (10–20% RMSE)	2 (11%)			6 (33%)		
Poor (>20% RMSE)	0 (0%)			2 (11%)		

Notes: ATT = Average Treatment Effect on the Treated, expressed as percentage change: $(\exp(\hat{\beta}) - 1) \times 100\%$. SE = Standard Error in log scale. Panel A standard errors are obtained from the wild bootstrap procedure in `multisynth`; 95% confidence intervals are constructed as $[\exp(\hat{\beta} \pm 1.96 \times \hat{\sigma}_{boot}) - 1] \times 100\%$. Panels B and C standard errors are the cross-period and cross-site standard deviations of weekly ATT estimates, respectively; confidence intervals follow the same transformation.

5.2.1 Main Treatment Effects

We implemented a partially pooled synthetic control method (Ben-Michael et al., 2022) to estimate the causal impact of the ULEZ expansion on NO_2 and NO_x concentrations at 18 monitoring stations in Greater London. The pooling parameter (ν) was empirically selected based on the balance possibility frontier, resulting in $\nu = 0.35$ for NO_2 and $\nu = 0.25$ for NO_x . Both models include unit fixed effects to account for time-invariant station-level heterogeneity, and outcomes are log-transformed to stabilize variance and interpret effects as percentage changes.

The synthetic control models achieve excellent pre-treatment fit for both pollutants. For NO_2 , the scaled global L_2 imbalance is 0.020 and the scaled individual L_2 imbalance is 0.103, both well below the recommended thresholds of 0.15 and 0.20, respectively (Ben-Michael et al. 2022). The pre-treatment RMSE is 0.0146 (1.46% relative error), and 89% of treated stations achieve good or excellent individual fit. For NO_x , the scaled imbalances are 0.022 (global) and 0.148 (individual), with a pre-treatment RMSE of 0.0211 (2.11%). Although slightly higher than NO_2 , these remain within acceptable ranges. The donor pool comprises 38 UK cities, with Plymouth, Coventry, Belfast, Leeds, and Nottingham contributing the largest weights.

During the 52-week post-implementation period, NO_2 concentrations declined on average by -5.4% (95% CI: -10.5% to $+0.3\%$), and NO_x by -6.4% (95% CI: -13.1% to $+1.2\%$). Both confidence intervals marginally include zero, indicating that while point estimates are consistently negative, statistical uncertainty remains non-trivial at the aggregate level.

The time-window analysis reveals that sub-period effects are more precisely estimated and consistently negative across all windows. For NO_2 , the effect exhibits a delayed pattern: short-term (0–15 weeks) reductions are modest at -3.9% (95% CI: -4.8% , -2.9%), mid-term (16–30 weeks) effects deepen to -6.5% (-7.4% , -5.6%), and long-term effects (31–52 weeks) stabilize at -5.5% (-6.3% , -4.7%). All three sub-period confidence intervals are entirely negative, in contrast to the overall estimate, reflecting that the aggregate effect is attenuated by the smaller short-term reduction.

For NO_x , an immediate strong reduction in -8.0% (-9.2% , -6.9%) is observed in the short term, sustained at -7.9% (-8.7% , -7.1%) in the medium term, before declining to -3.5% (-5.1% , -1.9%) in the long term.

Cohort-level analysis reveals heterogeneous spatial patterns in all three ULEZ expansion phases. For NO_2 , Inner London (-5.5% ; 95% CI: -9.3% , -1.4%) and Outer London (-5.5% ; 95% CI: -8.5% , -2.4%) exhibit confidence intervals completely below zero, while Central London (-4.5% ; -13.9% , $+5.9\%$) shows an interval spanning zero. For NO_x , the spatial pattern differs: Central London exhibits the largest reduction (-12.2% ; -17.9% , -6.0%) with a confidence interval entirely below zero, consistent with the high traffic density and elevated baseline NO_x concentrations at kerbside and urban traffic sites in this zone. In contrast, the inner London (-4.8% ; -11.6% , $+2.6\%$) and Outer London (-4.1% ; -8.1% , $+0.2\%$) intervals include zero, suggesting more modest and uncertain NO_x reductions in these cohorts.

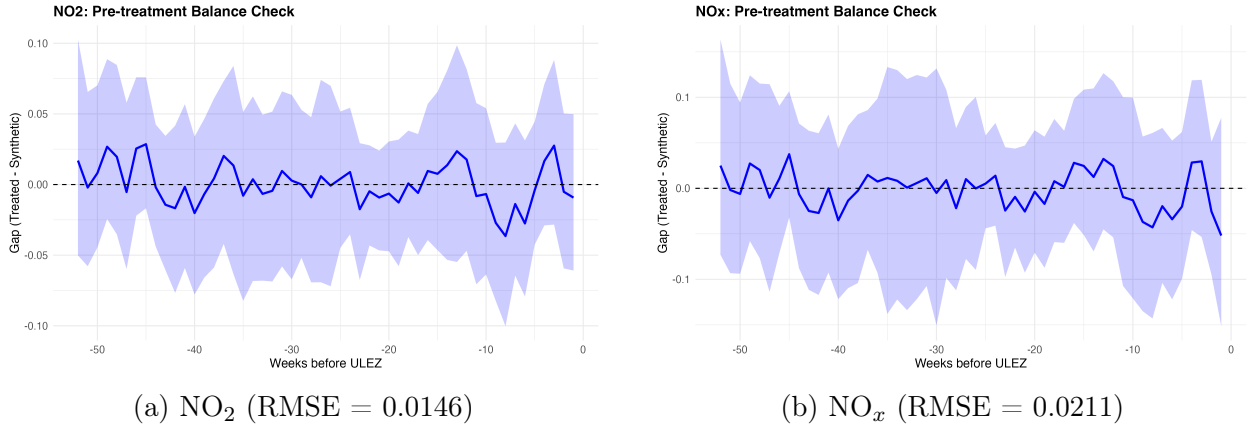


Figure 5: Pre-treatment balance check. Gap between treated and synthetic control units during the 52-week pre-intervention period. Values close to zero indicate good model fit. Shaded areas represent pointwise 95% confidence intervals.

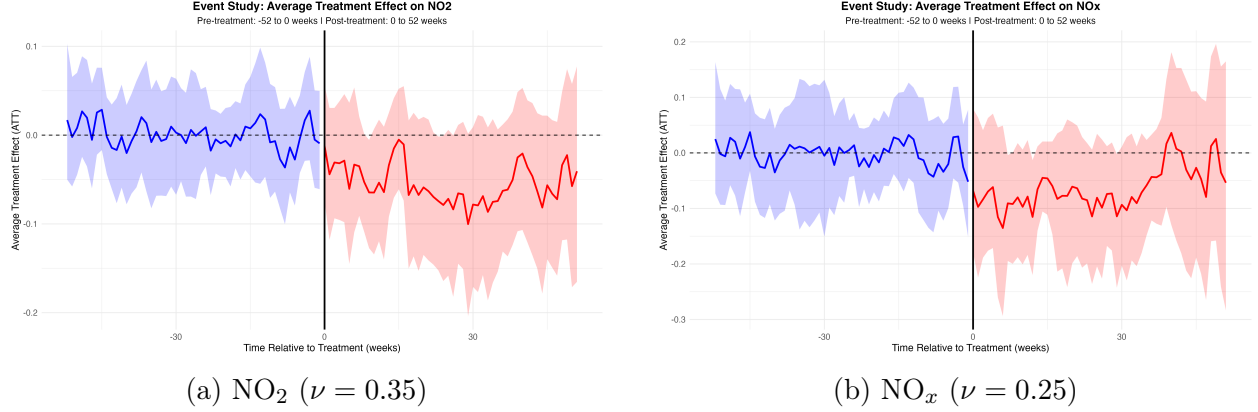


Figure 6: Event study estimates of ULEZ impacts on nitrogen oxide concentrations. Blue shading indicates pre-treatment period (-52 to 0 weeks); red shading indicates post-treatment period (0 to 52 weeks). Shaded areas represent pointwise 95% confidence intervals based on heteroskedasticity-robust standard errors. Vertical line at Time = 0 denotes policy implementation.

5.2.2 Robustness and Placebo Tests

To validate the causal interpretation of our findings, we conducted two robustness checks that examine spatial randomization and sample stability.

To verify that the estimated effects are not driven by random spatial patterns, we conduct a spatial (in-space) placebo test following a staggered permutation design. In each of 200 permutations, 18 control sites are randomly assigned to fake treatment cohorts mirroring the true staggered structure (4/7/7 sites at 2019/2021/2023 dates), and the model is re-estimated on this placebo dataset. Across all permutations, no placebo ATT approached the magnitude of the observed effects (NO_2 : true ATT = -0.054 log units, $p < 0.005$; NO_x : true ATT = 0.064 log units, $p < 0.005$), confirming that estimated reductions are unlikely to arise from random spatial variation.

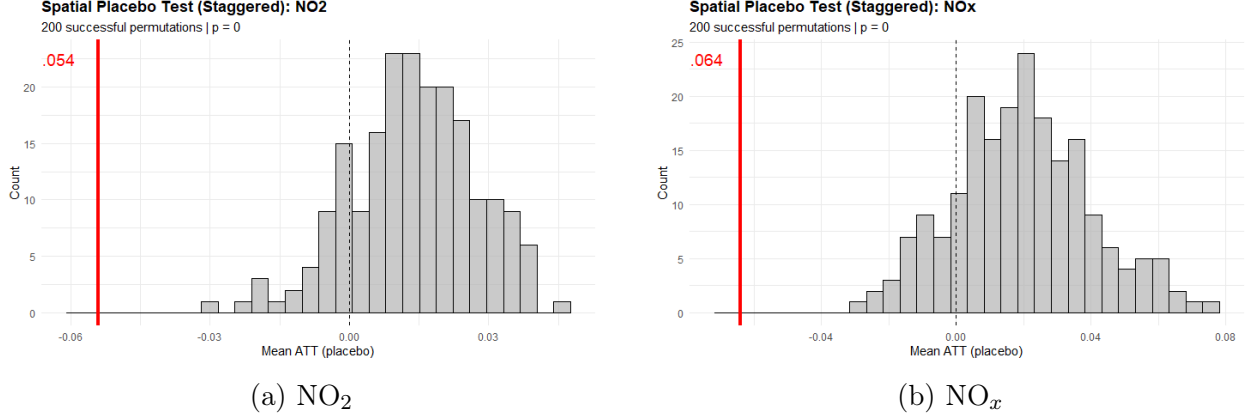


Figure 7: Spatial placebo test distributions (200 permutations). Red lines indicate the observed treatment effects (NO₂: -0.054 log units; NO_x: -0.064 log units).

Result stability is assessed through leave-one-out cross-validation, in which each of the 18 treated sites is sequentially removed and the model re-estimated on the remaining sample.

For NO₂, estimates range from -4.21% (MY1 removed) to -5.90% (LW4 removed), a spread of 1.69 percentage points around the full-sample estimate of -5.4% . MY1 (Marylebone Road, Westminster, Central London, 2019 cohort) produces the most attenuated estimate upon removal, implying a stronger-than-average individual treatment response consistent with its location on one of London’s busiest kerbside arterials. LW4 (Lewisham, Inner London, 2021 cohort) exerts the opposite influence: its removal pushes the aggregate estimate more negative, suggesting a weaker individual treatment response relative to the cross-site average.

For NO_x, the estimates range from -4.39% (WM6 removed) to -7.64% (LW4 removed), a spread of 3.25 percentage points around the full-sample estimate of -6.4% . A consistent spatial pattern emerges: WM6 (Oxford Street, Westminster, Central London, cohort 2019) mirrors the role of MY1 in NO₂ analysis, with its removal producing the most attenuated estimate and indicating an above-average response to treatment at this high-traffic urban location. LW4 again exerts a moderating influence on both pollutants, suggesting that its weaker treatment response reflects the different traffic composition and the lower baseline concentrations characteristic of its Inner London

location.

Across both pollutants, all 18 leave-one-out estimates remain negative with confidence intervals entirely below zero, confirming that the aggregate findings are robust and not driven by any single monitoring site.

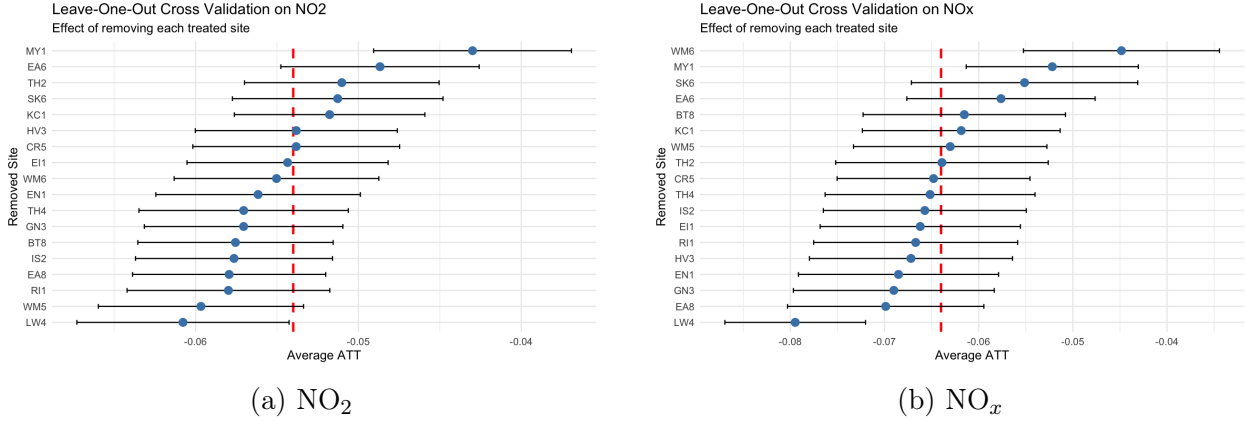


Figure 8: Leave-one-out cross-validation results. Red dashed lines indicate full model estimates. Error bars represent standard errors. The consistency of estimates across site exclusions demonstrates result stability.

6 Discussion and Conclusion

This study provides cautious but consistent evidence that ULEZ contributed to sustained reductions in nitrogen oxide pollution in London. Over the first 52 weeks post-treatment, NO₂ declined by -5.4% and NO_x by -6.4% relative to synthetic controls. While aggregate confidence intervals marginally include zero, sub-period estimates are consistently negative across all time windows, and both spatial placebo tests and leave-one-out cross-validation confirm that these reductions are unlikely to arise from random variation or the influence of any single monitoring site. From a methodological perspective, the integration of machine learning-based meteorological normalization and partially pooled synthetic controls sets a reproducible framework for future evaluations of urban air quality policies.

Although the direction and robustness of the effects are encouraging, aggregate confidence intervals including zero highlight the statistical challenges inherent in evalu-

ating environmental policies in real-world conditions. As more post-ULEZ data become available, future studies should revisit these estimates to assess longer-term trajectories and heterogeneity across pollutant types and spatial zones. In general, this research supports the ULEZ as a promising but not singularly decisive tool in the ongoing effort to improve urban air quality.

6.1 Discussion

This study provides a comprehensive causal evaluation of London’s Ultra Low Emission Zone (ULEZ) on NO_2 and NO_x concentrations using a partially pooled synthetic control framework with meteorological normalization. Our estimates indicate that ULEZ implementation was associated with persistent reductions in nitrogen oxides relative to synthetic controls. Over the first 52 weeks post-treatment, average reductions of -5.4% (95% CI: -10.5% , $+0.3\%$) for NO_2 and -6.4% (-13.1% , $+1.2\%$) for NO_x were observed. While both aggregate confidence intervals marginally include zero, sub-period estimates are more precisely identified and consistently negative across all time windows, strengthening the credibility of a treatment-related impact.

The temporal dynamics of the two pollutants reveal contrasting mechanisms. For NO_2 , effects follow a delayed accumulation pattern—modest in the short term (-3.9%), deepening in the mid-term (-6.5%), and stabilizing in the long term (-5.5%)—consistent with gradual fleet turnover, whereby non-compliant vehicles are progressively replaced or retired as owners respond to the daily charge. For NO_x , the pattern is reversed: a strong immediate effect (-8.0%) that is sustained in the mid-term (-7.9%) but diminishes in the long term (-3.5%). This divergence may reflect the fact that NO_x is more sensitive to immediate changes in diesel vehicle usage, while NO_2 reductions require a longer adjustment period tied to vehicle replacement cycles.

Cohort-level heterogeneity further illuminates the spatial dimension of policy effectiveness. For NO_2 , Inner and Outer London cohorts exhibit nearly identical and precisely estimated reductions (-5.5% each), while Central London’s estimate (-4.5%) carries substantial uncertainty, likely owing to the small cohort size of four stations and

a compressed pre-treatment window. For NO_x , the pattern reverses: Central London shows the largest and most precisely estimated reduction (-12.2%), consistent with the high traffic density and elevated baseline concentrations at kerbside and urban traffic monitoring sites in this zone. The more modest and uncertain NO_x reductions in Inner and Outer London may reflect lower baseline traffic intensity and a more heterogeneous vehicle fleet composition in these areas.

Interpretation of effect magnitudes and statistical uncertainty deserves careful consideration. The aggregate confidence intervals including zero at the 52-week horizon should not be interpreted as evidence of no effect. As Bonander et al. (2021) note, synthetic control applications in environmental epidemiology frequently encounter wide confidence intervals due to outcome volatility and the limited number of treated units available for inference. The consistency of negative point estimates across all sub-periods, cohorts, and robustness checks—including spatial placebo tests ($p < 0.005$ for both pollutants) and leave-one-out cross-validation in which all 18 site exclusions yield negative estimates with confidence intervals entirely below zero—provides strong indirect evidence that the observed reductions reflect genuine policy effects rather than statistical noise.

This analysis contributes methodologically by integrating machine learning-based weather normalization, partially pooled synthetic controls, and multiple robustness checks within a unified causal inference framework. Compared to previous evaluations of London’s ULEZ (Ma et al. 2021, Hajmohammadi and Heydecker 2022, Prieto-Rodríguez et al. 2022), this study offers extended post-policy follow-up covering all three expansion phases, explicit adjustment for meteorological confounding through random forest normalization, and a multi-pronged falsification strategy. These features address key limitations identified in the prior literature, including the absence of formal parallel trends validation and the restriction of analyses to the initial implementation of 2019.

6.2 COVID-19 as a Common Shock

A substantial concern is that our meteorological normalization model was trained on data spanning 2018–2024, including the COVID-19 pandemic period (March 2020–June 2021). During this time, traffic volumes and emissions patterns underwent unprecedented structural changes that were not related to either weather or ULEZ policy. This raises the question of whether the random forest model may have conflated pandemic-induced emission reductions with temporal or meteorological patterns, potentially biasing our normalized concentration estimates and, consequently, our synthetic control counterfactuals.

However, several theoretical and empirical features of our analysis suggest that this concern, while important, may not fatally compromise causal inference. First, the differencing logic inherent to synthetic control methods is specifically designed to handle common shocks—that is, unobserved confounders that affect both treated and control units in a similar way (Abadie et al. 2010, Abadie 2021). If the normalization model introduces systematic bias by misattributing COVID-related emissions to temporal patterns, this bias should affect London and control cities *symmetrically*, provided that both groups were subject to the same nationwide lockdown policies. In such cases, the synthetic control weights, which are calibrated to reproduce London’s pre-treatment trajectory using control cities, would implicitly replicate the COVID shock in the counterfactual, and the treatment effect estimate would reflect the difference *net of* the common pandemic impact.

Second, the UK implemented nationwide lockdown measures starting on March 23, 2020, affecting all cities in our sample simultaneously. Although behavioral responses and economic structures vary between regions, the first-order traffic and emissions shock was common, consistent with the assumption that COVID-19 functioned as a shared time effect rather than a unit-specific confounder. This is further supported by Google mobility data, which showed a 75% decrease in transit station activity and an overall 63% reduction in movement across the UK during lockdown periods (Sarran et al. 2020).

Third, and most critically, our spatial placebo tests provide strong empirical evi-

dence against the hypothesis that our results are driven by pandemic artifacts. Among 200 successful permutations in which control sites were randomly assigned to fake treatment cohorts that mirror the true staggered structure, no placebo ATT approached the magnitude of the observed effects (NO_2 : -5.4% ; NO_x : -6.4%). If COVID-induced normalization bias were systematically generating spurious treatment effects, we would expect placebo cities – all of which experienced the pandemic—to also exhibit comparable reductions when randomly designated as treated. The fact that only London shows a consistent post-ULEZ reduction suggests that our estimates capture policy-specific impacts rather than pandemic noise.

Fourth, the temporal dynamics of the treatment effects is inconsistent with a COVID-19 artifact. For NO_2 , the effects follow a delayed accumulation pattern: moderate in the short term (-3.9%), deepening in the mid-term (-6.5%) and stabilizing in the long term (-5.5%). For NO_x , the effects are immediate and sustained before gradually decreasing (-8.0% , -7.9% , -3.5% over short-, mid-, and long-term windows, respectively). Neither pattern exhibits sharp discontinuities or reversals aligned with the start and end dates of lockdown that would be expected if pandemic dynamics were driving the results. This is more consistent with gradual mechanisms such as fleet turnover and sustained behavioral adaptation.

Nevertheless, we cannot fully exclude the possibility of differential impacts of COVID between cities. London’s economic structure, higher reliance on public transport, and greater prevalence of remote work may have resulted in larger or more persistent traffic reductions compared to smaller UK cities, violating the symmetry assumption underlying our argument. If such an asymmetry exists, our estimates could conflate genuine ULEZ effects with London-specific pandemic dynamics. Ideally, future work should explore alternative normalization strategies – such as restricting training data to the pre-pandemic period (2018–February 2020), explicitly modeling lockdown periods as covariates, or employing segmented normalization for pre and post-pandemic eras – to quantify the sensitivity of our findings to this methodological choice.

6.3 Conclusion

This study provides cautious but methodologically rigorous evidence that London’s ULEZ contributed to sustained reductions in nitrogen oxide pollution, despite aggregate confidence intervals that marginally include zero and the confounding presence of the COVID-19 pandemic. The reliability of these findings is anchored in multiple layers of validation: spatial placebo tests rule out random geographic variation as an explanation, and leave-one-out cross-validation demonstrates result stability across site exclusions. Sub-period estimates are consistently negative across all time windows and cohorts, further reinforcing the credibility of a genuine policy effect.

The direction and robustness of our effect estimates—average reductions of -5.4% for NO_2 and -6.4% for NO_x over the first 52 weeks, consistently negative across all specifications and robustness checks—are encouraging and qualitatively align with descriptive evidence of fleet compliance and traffic reductions reported by Transport for London. However, the marginal statistical uncertainty at the aggregate level and the potential for pandemic-related confounding highlight the empirical challenges inherent in disentangling policy effects from concurrent macroeconomic shocks in real-world settings.

As additional post-ULEZ data accumulate and the pandemic’s disruptive effects recede, future research should revisit these estimates to assess longer-term trajectories, explore heterogeneity by neighborhood characteristics and pollution sources (e.g., roadside versus background sites), and quantify synergies with complementary policies such as Low Traffic Neighbourhoods and public transport investments.

In sum, this research supports the ULEZ as a *promising but not singularly decisive* tool in the ongoing effort to improve urban air quality. Although our estimates suggest that the policy has contributed to measurable, though modest, reductions in nitrogen oxides, achieving compliance with UK and EU air quality standards will likely require a portfolio of interventions—encompassing not only vehicle access restrictions but also accelerated electrification of public and private fleets, demand management measures, and continued investment in active transport infrastructure. The methodological

blueprint developed here—combining data-driven normalization, causal inference, and extensive falsification—offers a path forward for a credible and reproducible evaluation of such policies in London and cities worldwide.

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